

# Signed transport ledger, sharpened Besov payment, and shifted-resonance boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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## 1 Purpose and scope

R-075 reconstructed the complete invariant-current Taylor chart, proved the principal second-order one-form, and identified a third-order transported tail. It then used

$$\|A^3 DA\|_{B_{1,1}^s} \lesssim \|A\|_{H^2} \|A\|_6^3, \quad s = \frac{1}{2} + \kappa, \quad (1.1)$$

which gives  $X^{1/2}Y^{1/2}$  for  $X = \|A\|_{H^2}^2$  and  $Y = \|A\|_6^6$ . That particular estimate has zero Young slack. The conclusion recorded in R-075 was too broad: (1.1) is not the sharp mixed  $H^2$ – $L^6$  estimate.

This note corrects that diagnosis. For  $0 < s < 1$ ,

$$\boxed{\|A^3 DA\|_{B_{1,1}^s} \lesssim_s \|A\|_{H^2}^{(1+s)/2} \|A\|_6^{(7-s)/2}.} \quad (1.2)$$

At the audited  $\kappa = 1/10$ , hence  $s = 3/5$ , the associated pairing has positive Young slack  $1/15$ . Any already constructed control-independent  $L^{15}(\Omega; C^{-3/5})$  current can therefore be paid with arbitrary Cameron–Martin and terminal-sextic allocations. This closes the base-frozen cubic one-form inside the R-075 transported tail and both nonresonant shifted-multiplier paraproducts.

The full signed endpoint still does not close. The remaining analytic object is the high–high resonance of the shifted coefficient with the uncontrolled geometric current. It cannot first be converted into an  $A$ -uniform multiplier norm. Nor is the complete physical affine-current Bregman remainder positive: an exact production fixture has negative curvature four times the retained square. The residual resonance must be kept with the terminal square, coefficient-curvature/Wick block, every R-063 lower chaos, R-066 trace transport, and the finite-low boundary in a single causal signed argument.

Thus this is a scoped T4 reduction and correction, not a tier promotion. It does not prove the paired adapted shifted enhancement, the signed endpoint lower bound, controlled-shell one-use, the  $q = 10/9$  Nelson theorem, an interacting measure, floor or regulator removal, or infinite volume. All statements use the fixed positive production floor, common real-even regulators, exact non-aliased products, and the R-066 finite-low convention.

## 2 The nonduplicating signed endpoint ledger

Suppress the generator and spatial sums. Put

$$M_0 = M(U), \quad M_1 = M(U + A), \quad D = DM(U)[A], \quad (2.1)$$

$$E = M_1 - M_0 - D = E_2 + E_3, \quad E_2 = \frac{1}{2} D^2 M(U)[A, A], \quad (2.2)$$

$$E_3 = \frac{1}{2} \int_0^1 (1-t)^2 D^3 M(U+tA)[A, A, A], dt. \quad (2.3)$$

For  $G = DU$ ,  $C = DA$ , define

$$w = M_0^T G, \quad \Delta w = (D + E)^T G, \quad c = M_1^T C, \quad (2.4)$$

and

$$F_A = B(U + A) - B(U) - DB(U)[A], \quad Q_J = G \otimes G - \Gamma_J. \quad (2.5)$$

After the R-075 principal  $E_2$ - $C$  one-form has been separated, the exact remaining endpoint block is

$$\Xi_J = \langle w, Q_{II} E_3^T C \rangle + \frac{1}{2} \langle F_A, Q_J \rangle + \langle \Delta w, Q_{II} c \rangle + \frac{1}{2} |c|_{Q_{II}}^2. \quad (2.6)$$

Because

$$\frac{1}{2} F_A : (G \otimes G) = \langle w, Q_{II} E^T G \rangle + \frac{1}{2} |\Delta w|_{Q_{II}}^2, \quad (2.7)$$

(2.6) reassembles without duplication as

$$\boxed{\Xi_J = \langle w, Q_{II} E_2^T G \rangle + \langle w, Q_{II} E_3^T (G + C) \rangle + \frac{1}{2} |\Delta w + c|_{Q_{II}}^2 - \frac{1}{2} F_A : \Gamma_J.} \quad (2.8)$$

The full R-075 remainder is the R-071 linear  $D$ - $C$  term, the principal  $E_2$ - $C$  term, and (2.8). Thus every diagonal and off-diagonal current appears once, the full terminal current square remains positive, and the coefficient-curvature/Wick channel remains signed.

For a deterministic or conditionally legitimate frozen shift, the coefficient term in (2.6) is one half of the complete R-063 chart:

$$\begin{aligned} & \sum_j \{F_A(x_j) : Q_j + DF_A(x_j) : (P_3 + P_1) \\ & + \frac{1}{2} D^2 F_A(x_j) : (P_4 + P_2^{\text{cross}} + \Sigma_j Q_j + P_0^{\text{cross}}) + \mathcal{R}_{3,j}^{F_A} : Q_j\}. \end{aligned} \quad (2.9)$$

Equation (2.9) is an equality ledger, not a positivity statement. In particular,  $\Sigma Q$  and the lower chaoses may not be merged into the trace a second time.

### 3 A sharpened cubic Besov theorem

#### 3.1 Statement and proof

Let  $0 < s < 1$  and  $A \in H^2(\mathbb{T}_L^3) \cap L^6$ . Then (1.2) holds. The estimate is tensor-entrywise, so finite component contractions change only its constant.

Let  $\Delta_j$  be inhomogeneous Littlewood–Paley blocks. Expand the four inputs and assign each monomial to its largest *input* dyadic index  $j$ , resolving ties by a fixed ordering. Write the resulting sum as

$$A^3 DA = \sum_{j \geq -1} G_j. \quad (3.1)$$

Every  $G_j$  has Fourier support in a ball of radius  $C2^j$ , even when two high inputs produce a low output. Hence  $\|G_j\|_{B_{1,1}^s} \lesssim 2^{sj} \|G_j\|_1$ . If the differentiated input is maximal, put  $\Delta_j DA$  in  $L^2$  and the other three partial sums in  $L^6$ . If an undifferentiated input is maximal, put that  $\Delta_j A$  in  $L^2$  and use  $\|DS_j A\|_6 \lesssim 2^j \|A\|_6$  for the differentiated partial sum. The finitely many tie patterns obey the same estimate, so

$$\|G_j\|_{B_{1,1}^s} \lesssim 2^{(1+s)j} \|\Delta_j A\|_2 \|A\|_6^3. \quad (3.2)$$

This input-maximum decomposition, rather than an output projection, retains all high-high-to-low interactions. Summing (3.2) gives the multilinear paraproduct bound

$$\|A^3 DA\|_{B_{1,1}^s} \lesssim \|A\|_{B_{2,1}^{1+s}} \|A\|_6^3 \quad (3.3)$$

with no frequency-separation hypothesis.

It remains to interpolate the single-input Besov norm. Split its dyadic sum, now legitimately by the input index, at  $K$ . Cauchy–Schwarz gives

$$\sum_{j \leq K} 2^{(1+s)j} \|\Delta_j A\|_2 \lesssim 2^{(1+s)K} \|A\|_2,$$

whereas

$$\sum_{j > K} 2^{(1+s)j} \|\Delta_j A\|_2 \lesssim 2^{-(1-s)K} \|A\|_{H^2}.$$

Balancing  $2^{2K} \simeq \|A\|_{H^2} / \|A\|_2$  proves

$$\|A\|_{B_{2,1}^{1+s}} \lesssim \|A\|_{H^2}^{(1+s)/2} \|A\|_2^{(1-s)/2} \quad (3.4)$$

and (1.2) follows from  $\|A\|_2 \lesssim \|A\|_6$  on the fixed torus. The inhomogeneous bottom block and the zero cases are absorbed into the fixed-volume constant.

### 3.2 The exact Young ledger

Let  $R \geq 0$  be the  $C^{-s}$  norm of a reconstructed current. From (1.2),

$$R\|A^3 DA\|_{B_{1,1}^s} \lesssim R X^a Y^b, \quad a = \frac{1+s}{4}, \quad b = \frac{7-s}{12}. \quad (3.5)$$

The unused exponent is

$$c = 1 - a - b = \frac{1-s}{6} > 0. \quad (3.6)$$

Weighted AM–GM therefore yields, for every  $\eta, \zeta > 0$ ,

$$\boxed{R X^a Y^b \leq \eta X + \zeta Y + C_s \eta^{-a/c} \zeta^{-b/c} R^{1/c}.} \quad (3.7)$$

The required random moment is  $6/(1-s)$ . At  $\kappa = 1/10$ ,

$$s = \frac{3}{5}, \quad a = \frac{2}{5}, \quad b = \frac{8}{15}, \quad c = \frac{1}{15}, \quad (3.8)$$

and hence

$$\boxed{R X^{2/5} Y^{8/15} \leq \eta X + \zeta Y + C \eta^{-6} \zeta^{-8} R^{15}.} \quad (3.9)$$

R-050 and R-063 supply every finite moment for their unshifted enhanced currents, so the fifteenth moment in (3.9) is available.

## 4 What the correction closes

Let

$$J = M(U)^T DU, \quad H_t = D^3 M(U + tA). \quad (4.1)$$

At fixed positive floor,  $D^3 M$  is globally bounded. The R-050/R-071 one-form argument gives  $J \in L^p(\Omega; C^{-s})$  for every finite  $p$  and the same is true after multiplication by a smooth coefficient depending only on  $U$ .

First freeze the third derivative at the uncontrolled field. The base-frozen cubic branch is

$$\mathcal{N}_{3,0} = \frac{1}{6} \left\langle M(U)^T DU, Q_{\Pi}[D^3 M(U)[A, A, A]^T DA \right\rangle. \quad (4.2)$$

Its coefficient is a finite smooth geometric one-form in  $U$ , independent of the chosen control in the norm estimate. It has every finite  $L^p(C^{-s})$  moment, so (3.7), and in particular (3.9), closes (4.2). Here “independent of the control” does not mean probabilistic independence; it means that the random norm and its moment bound do not grow with  $A$ .

There is a second useful reduction. At a smooth cutoff, use Bony’s split

$$H_t J = T_{H_t} J + T_J H_t + H_t \circ J. \quad (4.3)$$

For  $s > 0$ , both paraproducts need only the boundedness of  $H_t$ :

$$\|T_{H_t} J\|_{C^{-s}} + \|T_J H_t\|_{C^{-s}} \lesssim \|H_t\|_{\infty} \|J\|_{C^{-s}}. \quad (4.4)$$

For the second estimate,  $\|S_{j-1} J\|_{\infty} \lesssim 2^{sj} \|J\|_{C^{-s}}$  and  $\|\Delta_j H_t\|_{\infty} \lesssim \|H_t\|_{\infty}$ . Thus (3.7) closes both nonresonant branches with a control-independent random constant.

Write

$$H_t = H_0 + K_t, \quad K_t = D^3 M(U + tA) - D^3 M(U). \quad (4.5)$$

The resonance  $H_0 \circ J$  is part of the existing R-050/R-063 smooth geometric enhancement and is also closed by (3.7). Consequently the only new rough multiplier product in the R-075  $E_3$ - $C$  tail is

$$\boxed{K_t \circ J}. \quad (4.6)$$

This statement is an algebraic/analytic branch reduction. It does not yet construct (4.6) for an arbitrary adapted control or prove its signed lower bound.

## 5 Why the shifted resonance cannot be separated

Boundedness of  $D^3 M$  does not make multiplication by  $H_t$  continuous on  $C^{-s}$ . The obstruction occurs in the actual production frame. On the active-real  $\sigma_3$  radial line, its nonlinear column is

$$v(r) = \frac{2er}{r^2 + e}, \quad v^{(4)}(\sqrt{e}) = -\frac{6}{e^{3/2}} \neq 0. \quad (5.1)$$

Take on a one-dimensional spatial carrier

$$U = \sqrt{e}, \quad A_N = \delta \cos(Nx), \quad J_N = N^s \cos(Nx), \quad (5.2)$$

with  $0 < \delta \ll \sqrt{e}$ , and fix  $t \in (0, 1]$ . Then  $\|J_N\|_{C^{-s}} \asymp 1$ , whereas

$$J_N \{v'''(U + tA_N) - v'''(U)\} \quad (5.3)$$

has zero Fourier mode  $\frac{1}{2}tv^{(4)}(\sqrt{e})\delta N^s + O(t^2\delta^2 N^s)$ . For each fixed  $t > 0$  this is comparable to  $\delta N^s$  when  $\delta$  is small. At  $t = 0$  the difference is identically zero, as it must be. Hence no  $A$ -uniform separated estimate for a nontrivial path time

$$\|K_t J\|_{C^{-s}} \leq C(\omega) \quad (5.4)$$

can follow from  $\|K_t\|_{\infty}$  and  $\|J\|_{C^{-s}}$ .

Moreover  $\|A_N\|_{H^2} \asymp \delta N^2$ , so the separated multiplier loss behaves like  $\|A_N\|_{H^2}^{s/2}$ . Appending this loss to (3.5) changes the total deterministic exponent from  $(5 + s)/6$  to

$$\frac{5 + s}{6} + \frac{s}{4} = \frac{10 + 5s}{12}. \quad (5.5)$$

At  $s = 3/5$ , (5.5) is  $13/12 > 1$ . The separated-multiplier architecture is therefore supercritical even though the corrected bare payload is not. This is a method counterexample, not a counterexample to the complete paired endpoint (2.8).

Taylor iteration does not repair the problem. Exactly,

$$E_3 = \frac{1}{6} D^3 M(U)[A, A, A] + E_4, \quad (5.6)$$

$$E_4 = \frac{1}{6} \int_0^1 (1-r)^3 D^4 M(U+rA)[A, A, A, A], dr. \quad (5.7)$$

The  $C$ -branch of (5.7) has the payload  $A^4 DA$ ; assigning one factor to  $H^2$  and four to  $L^6$  already gives  $1/2 + 4/6 = 7/6 > 1$  in spatial Holder. Thus an infinite base-point Taylor expansion worsens the deterministic budget.

The correct object to retain is the exact paired remainder

$$\mathcal{E}_3(U, A) = M(U + A) - M(U) - DM(U)[A] - \frac{1}{2} D^2 M(U)[A, A]. \quad (5.8)$$

At fixed floor,

$$|\mathcal{E}_3(U, A)| \lesssim_e |A|^3, \quad |D_A \mathcal{E}_3(U, A)| \lesssim_e |A|^2, \quad (5.9)$$

and, schematically,

$$\begin{aligned} |\delta \mathcal{E}_3| &\lesssim_e (|A|^2 + |A'|^2) |\delta A| \\ &\quad + (|A|^3 + |A'|^3) |\delta U|. \end{aligned} \quad (5.10)$$

The explicit control-high part remains degree four. The second line of (5.10), in which rough  $U$  enters an adapted shifted coefficient, is exactly the missing paired enhancement rather than an extra deterministic power.

## 6 Affine Bregman and path positivity are false shortcuts

For

$$F(z, y) = \frac{1}{2} \sum_r |M_r(z)^T y|_{Q_{\Pi}}^2, \quad (6.1)$$

put  $z_1 = z + A$ ,  $y_1 = y + C$ ,  $w_0 = M(z)^T y$ , and

$$\delta w = M(z_1)^T y_1 - M(z)^T y, \quad \ell = DM(z)[A]^T y + M(z)^T C. \quad (6.2)$$

The remainder after both Frechet first variations is exactly

$$F(z_1, y_1) - F(z, y) - DF(z, y)[A, C] = \frac{1}{2} |\delta w|_Q^2 + \langle w_0, Q(\delta w - \ell) \rangle. \quad (6.3)$$

Although (6.3) is an affine Bregman remainder,  $F$  is not convex in the physical field variables.

Take

$$z = e_1 + e_3, \quad A = e_1 - e_3, \quad y = e_3, \quad C = 0, \quad d = 2 + e. \quad (6.4)$$

Only the  $r = 3$  nonlinear current survives, and

$$w_0 = (0, -2/d), \quad w_1 = 0, \quad \ell = (0, -2/d). \quad (6.5)$$

With  $q_{22} = 3/(320P)$ , the two terms in (6.3) are

$$\frac{1}{2} |\delta w|_Q^2 = \frac{2q_{22}}{d^2}, \quad \langle w_0, Q(\delta w - \ell) \rangle = -\frac{8q_{22}}{d^2}, \quad (6.6)$$

The signed remainder is therefore

$$D_F = -\frac{6q_{22}}{d^2} = -\frac{9}{160P(2+e)^2} = -0.0035156249999956\dots \quad (6.7)$$

The curvature is negative four times the retained square, and the full remainder is negative three times the base raw energy. More generally, if the endpoint is  $ce_1$ , then

$$\frac{D_F(c)}{F_0} = -\frac{4c(e+1) - 5e - 2}{e+2} \longrightarrow -\infty. \quad (6.8)$$

This rules out convex Bregman, square-only, and fixed-multiple-of-base-energy shortcuts. Since  $|A|^2 \sim c^2$ , it does not refute a Cameron–Martin–cost lower bound.

The exact affine path makes the same point. Let

$$Z_t = U + tA, \quad Y_t = G + tC, \quad w_r(t) = M_r(Z_t)^T Y_t. \quad (6.9)$$

Then

$$\dot{w}_r = DM_r(Z_t)[A]^T Y_t + M_r(Z_t)^T C, \quad (6.10)$$

$$\ddot{w}_r = D^2 M_r(Z_t)[A, A]^T Y_t + 2DM_r(Z_t)[A]^T C, \quad (6.11)$$

and

$$F(1) - F(0) - F'(0) = \int_0^1 (1-t) \sum_r \{|\dot{w}_r|_Q^2 + \langle w_r, Q\ddot{w}_r \rangle\} dt. \quad (6.12)$$

At zero floor, the fixture (6.4) has the normalized current

$$k(t) = \frac{(t-1)(t+1)^2}{1+t^2}. \quad (6.13)$$

Its positive and curvature integrals are exactly

$$\int_0^1 (1-t)(k')^2 dt = \frac{7}{2} - \log 4 - \frac{\pi}{2} = 0.5429093120\dots, \quad (6.14)$$

$$\int_0^1 (1-t)kk'' dt = -5 + \log 4 + \frac{\pi}{2} = -2.0429093120\dots \quad (6.15)$$

Their sum is  $-3/2$ . Wick ordering subtracts the matching second variation  $D^2 B(Z_t)[A, A] : \Gamma/2$  and supplies no new sign.

There is also a finite-dimensional adapted centering diagnostic. If a strict-past scalar  $X \sim N(0, 1)$  supplies  $G = Xe_3$ , the control (6.4) is activated on  $\{|X| > a\}$ , and  $\Gamma = e_3 e_3^T$ , then

$$\mathbb{E} D_{F_\Gamma} = -\frac{6q_{22}}{d^2} \mathbb{E}[(X^2 - 1)\mathbf{1}_{\{|X| > a\}}] = -\frac{6q_{22}}{d^2} 2a\varphi(a) < 0. \quad (6.16)$$

At  $a = 1$  the value is  $-0.0017013566567\dots$ ; smooth selectors preserve the sign. This is a causal finite-dimensional diagnostic, not a complete spatial one-use counterexample.

## 7 The exact causal frontier

The terminal shift cannot simply be frozen at every Gaussian root. If  $g_1, g_2$  are successive fresh shells and the strict-past second control is  $a_2 = f(g_1)$ , then  $a_2$  is admissible at shell two but the terminal shift  $A = a_2$  is not measurable before root one. R-075's smooth infinite-Hermite example makes this failure explicit. A valid proof must assign every monomial to its largest causal root rather than condition on the terminal coefficient wholesale.

The largest-root audit suggests the following three candidate classes. It is a causal decomposition proposal; this note does not prove the uniform root expansion, conditional moment, or shell summability for the first two classes.

1. If a fresh Gaussian root is largest, all earlier and equal-shell strict-past controls may be conditionally frozen at that root. A successor must still prove that the exact R-063  $P_3/P_1$  and  $P_4/P_2/\Sigma Q/P_0$  compensators reassemble and sum uniformly after this conditioning.
2. If a largest control pair lies among the explicit  $A^3DA$  factors, the candidate move is to charge that pair first to its own shell Cameron–Martin energy and use the remaining positive slack. The no-reuse shell charge and the residual dyadic summation are not proved here.
3. The hard branch has its largest control roots only inside the shifted coefficient  $K_t$ , while the explicit  $A^3DA$  payload is lower. Absolute fractional multiplier estimates are incompatible with both shell summability and Young slack for  $s > 1/2$ . This coefficient-only balanced-high branch must remain signed and cancel against the terminal square, coefficient curvature, R-063 lower chaoses, and trace transport.

The smallest honest successor is therefore a three-class largest-root proposition. Its apparently hardest component is a *coefficient-only balanced-root signed Carleson lemma*: after the R-073 endpoint is reassembled exactly, the complete block containing (4.6), the corresponding terminal-square polarization, the coefficient Wick forest, R-066 trace, and finite-low term would have to obey

$$\mathbb{E}\mathcal{B}_J(A) \geq -\eta \mathbb{E}X - \zeta \mathbb{E}Y - C_{\eta,\zeta} \quad (7.1)$$

uniformly in  $J$ . The notation  $\mathcal{B}_J$  refers to that complete nonduplicating block, not to the isolated resonance. Only after the fresh-root, explicit-payload-high, and coefficient-only estimates are all proved for regular strict-past controls may one invoke R-075 graph recovery. The remaining R-066 reductions would then give controlled-shell one-use, after which the already established  $q = 10/9$  equivalence returns the proof to the Nelson synthesis.

## 8 Evidence and failure map

The proof route after R-076 is:

1. **Exact ledger closed:** (2.8) puts the second-order curvature, third-order transport, terminal square, and Wick trace into one nonduplicating endpoint identity.
2. **Previous diagnosis corrected:** R-075's coarse estimate (1.1) is Young-critical, but the sharp mixed estimate (1.2) has slack  $(1 - s)/6$ . The old negative result is retained historically and narrowed; the overbroad conclusion is recorded as AUDIT-2026-07-25-A13-R075-COARSE-TRANSPORT-CRITICALITY.
3. **Closed analytically:** the base-frozen cubic one-form, both nonresonant paraproducts, and the unshifted resonance are paid by (3.7). At  $s = 3/5$  the required moment is fifteen.
4. **Closed negatively:** the physical affine Bregman remainder and its path-square completion are not positive, by (6.7) and (6.14)–(6.15).
5. **Closed negatively:** a separated shifted-multiplier norm is false by the zero-mode fixture (5.1)–(5.5). These two shortcut failures are registered together as NG-2026-07-25-A13-BREGMAN-AND-SEPARATED-SHIFTED-MULTIPLIER.
6. **Still open:** prove the three proposed largest-root branch estimates. The fresh-root and explicit-payload-high branches need their own uniform expansion and summability lemmas; the coefficient-only balanced-high branch must remain with the complete lower-chaos, square, trace, and finite-low channels and is conjectured to satisfy (7.1).

No route inspected in this note is left implicit. A base-point Taylor iteration fails by the  $A^4DA$  spatial budget; terminal root freezing fails causally; affine and Wick Bregman positivity fail on an exact fixture; separated multiplier control is supercritical; and the three largest-root branches are a recorded successor decomposition rather than proved branch closures.

## 9 Devil’s-advocate review

1. **“R-075’s third-order no-go remains valid without change.” UPHELD only for its coarse estimate, rejected as a general no-go.** Equation (1.2) leaves positive slack. The registry wording is corrected rather than silently deleted.
2. **“The fifteenth moment closes the entire shifted tail.” UPHELD as false.** It closes only currents whose norm is independent of the chosen control, plus the two nonresonant paraproducts. The shifted resonance (4.6) is not covered.
3. **“Bounded  $D^3M$  is an  $A$ -uniform multiplier theorem.” UPHELD as false.** Equal high frequencies in (5.2)–(5.3) leak a zero mode of size  $\delta N^s$ .
4. **“Taylor expansion around  $U$  removes the shifted coefficient.” UPHELD as false.** The next exact remainder carries  $A^4DA$  and is spatially supercritical by (5.7).
5. **“The retained path square makes the affine remainder positive.” UPHELD as false.** In (6.6) curvature is negative four times the square.
6. **“Wick subtraction repairs affine convexity.” UPHELD as false.** Equation (6.16) has strictly negative centered expectation; Wick ordering supplies the trace identity, not a sign.
7. **“Every strict-past terminal control can be frozen before every root.” UPHELD as false.** Later controls may depend on an earlier Gaussian root. Largest-root causal assignment is necessary.
8. **“The fresh-root and explicit-payload-high classes are already closed.” UPHELD as false.** This note identifies the conditional freezing and shell-harvest candidates but proves neither the complete root expansion nor a cutoff-uniform dyadic summation for those classes.
9. **“The multiplier fixture disproves one-use.” DISMISSED.** It disproves only the separated norm architecture. The complete paired endpoint may cancel through channels absent from the fixture.
10. **“The affine Bregman fixture disproves a Cameron–Martin bound.” DISMISSED.** Its large endpoint family has quadratic control cost. It excludes convexity and repeated base-energy payment, not the requested one-use inequality.
11. **“The zero-mode multiplier witness is nonzero also at  $t = 0$ .” UPHELD as false.** Since  $K_0 = 0$ , the witness vanishes at the initial path time. Equation (5.3) fixes  $t \in (0, 1]$  and displays the leading factor  $t/2$ ; the separated-multiplier no-go survives at every fixed nontrivial path time.
12. **“This result closes Sector A.” UPHELD as an overclaim.** The signed paired resonance, one-use, and Nelson theorem remain open; the tier stays T4.



## 10 Reproduction and independent audit

The primary executable derives every production coefficient from A1, checks the signed endpoint identity on random six-real fixtures, derives the exact Besov/Young exponents, evaluates the affine and centered selector fixtures, integrates the path square and curvature, and verifies the radial multiplier leakage. The non-importing executable rebuilds the complex Pauli frames, uses finite differences for the endpoint ledger, composite Simpson quadrature for the path, and an automatic Taylor-jet algebra for the radial derivatives.

```
python codes/foundations/  
a13_classii_signed_transport_besov_bregman_resonance_verify.py
```

### Result footer

Result ID: A13-CLASSII-SIGNED-TRANSPORT-BESOV-BREGMAN-RESONANCE-REDUCTION

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Precise statement:      | The exact signed endpoint has the nonduplicating form (2.8). The sharp cubic Besov payload has Young slack $(1-s)/6$ , closing the control-independent cubic one-form and both nonresonant paraproduts. The only new rough product in that analytic split is the paired adapted shifted high-high resonance; affine-Bregman and separated multiplier shortcuts are false. No causal largest-root branch is closed. |
| Scope:                  | Fixed-floor A1/R-050/R-063/R-066/R-071/R-073/R-075; common real-even cutoffs; regular whole-shell strict-past controls; finite-low retained.                                                                                                                                                                                                                                                                       |
| Dependencies:           | A1; R-050; R-063; R-066; R-071; R-073--R-075.                                                                                                                                                                                                                                                                                                                                                                      |
| Evidence grade:         | ANALYTIC, EXACT, EXECUTED.                                                                                                                                                                                                                                                                                                                                                                                         |
| Reproduction command:   | python codes/foundations/<br>a13_classii_signed_transport_besov_bregman_resonance_verify.py                                                                                                                                                                                                                                                                                                                        |
| Expected output:        | primary + independent + integrated aggregate PASS.                                                                                                                                                                                                                                                                                                                                                                 |
| Falsification gate:     | A signed-ledger, exponent, Bregman, path, multiplier, hash, PDF, scope, or release assertion fails.                                                                                                                                                                                                                                                                                                                |
| Tier before / after:    | T4 / T4; no promotion.                                                                                                                                                                                                                                                                                                                                                                                             |
| No-overclaim statement: | No largest-root branch theorem, paired adapted shifted enhancement, complete signed endpoint lower bound, one-use, Nelson, measure, removal, infinite volume, or T5--T7.                                                                                                                                                                                                                                           |
| Next required action:   | Prove the fresh-root and explicit-payload-high branch lemmas and the coefficient-only balanced-root signed Carleson component with every R-063 lower chaos, terminal square, R-066 trace, and finite-low boundary retained.                                                                                                                                                                                        |